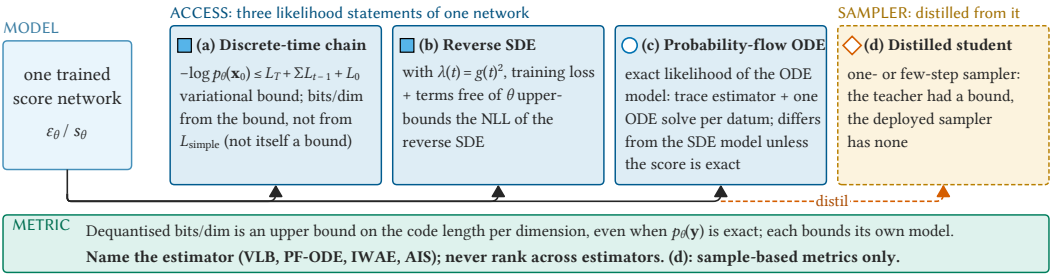
MODEL
one trained score network
ε_θ / s_θ
ACCESS: three likelihood statements of one network
(a) Discrete-time chain
$-\log p_\theta(\mathbf{x}_0) \le L_T + \Sigma L_{t-1} + L_0$ variational bound; bits/dim from the bound, not from L_{simple} (not itself a bound)
(b) Reverse SDE
with $\lambda(t) = g(t)^2$, training loss + terms free of θ upper-bounds the NLL of the reverse SDE
(c) Probability-flow ODE
exact likelihood of the ODE model: trace estimator + one ODE solve per datum; differs from the SDE model unless the score is exact
SAMPLER: distilled from it
(d) Distilled student
one- or few-step sampler: the teacher had a bound, the deployed sampler has none
distil
METRIC
Dequantised bits/dim is an upper bound on the code length per dimension, even when $p_\theta(\mathbf{y})$ is exact; each bounds its own model.
Name the estimator (VLB, PF-ODE, IWAE, AIS); never rank across estimators. (d): sample-based metrics only.